

Class 12: Introduction to Genome Informatics Lab

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Section 1: Identify Genetic Variants of Interest

Q1: What are those 4 candidate SNPs?

Based on my OMIM search, rs12936231, rs8067378, rs9303277, and rs7216389 are the 4 candidate SNPs.

Q2: What three genes do these variants overlap or effect?

ZPBP2, GSDMB, ORMDL3.

Q3: What is the location of rs8067378 and what are the different alleles for rs8067378?

Location: Chromosome 17:39895095 (forward strand) Alleles: A/C/G Ancestral: G Highest population MAF: 0.50

Q4: Name at least 3 downstream genes for rs8067378?

3 genes downstream of ZPBP2 are GSDMB, ORMDL3, and LRRC3C.

Q5: What proportion of the Mexican Ancestry in Los Angeles sample population (MXL) are homozygous for the asthma associated SNP (G|G)?

Downloaded a CSV file from Ensemble < https://useast.ensembl.org/Homo_sapiens/Variation/Sample?db=core;v=rs8067378;vdb=variation;vf=959672880#373531_tablePanel >

Here we read this CSV file

```
mxl <- read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(mxl)
```

	Sample..	Male.	Female.	Unknown.	Genotype..	forward.	strand.	Population.s.	Father
1					NA19648	(F)		A A ALL, AMR, MXL	-
2					NA19649	(M)		G G ALL, AMR, MXL	-
3					NA19651	(F)		A A ALL, AMR, MXL	-
4					NA19652	(M)		G G ALL, AMR, MXL	-
5					NA19654	(F)		G G ALL, AMR, MXL	-
6					NA19655	(M)		A G ALL, AMR, MXL	-
	Mother								
1		-							
2		-							
3		-							
4		-							
5		-							
6		-							

```
table(mx1$Genotype..forward.strand.)
```

```
A|A  A|G  G|A  G|G
22  21  12   9
```

```
table(mx1$Genotype..forward.strand.) / nrow(mx1) * 100
```

```
      A|A      A|G      G|A      G|G
34.3750 32.8125 18.7500 14.0625
```

~14% of the Mexican Ancestry in Los Angeles sample population are homozygous for the asthma associated SNP.

Now, let's look at a different population. Proffessor Grant picked GBR!

```
gbr <- read.csv("373522-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
```

Find proportion of G|G.

```
round(table(gbr$Genotype..forward.strand.) / nrow(gbr) * 100, 2)
```

```
      A|A      A|G      G|A      G|G
25.27 18.68 26.37 29.67
```

~30% of the British in England and Scotland population is homozygous for the asthma associated SNP.

This variant that is associated with childhood asthma is more frequent in the GBR population than the MXL population.

Let's now dig into this further!

Q6. Back on the ENSEMBLE page, use the “search for a sample” field above to find the particular sample HG00109. This is a male from the GBR population group. What is the genotype for this sample?

The genotype for this sample is G|G.

Section 2: Initial RNA-Seq analysis

Q7: How many sequences are there in the first file? What is the file size and format of the data? Make sure the format is fastqsanger here!

There are 3,863 sequences in the first file. The file size is 757 KB and the data is in a fastqsanger format.

Q8: What is the GC content and sequence length of the second fastq file?

The second data set has a GC content of 54%, and a 50-75 sequence length.

Q9: How about per base sequence quality? Does any base have a mean quality score below 20?

The per base sequence quality score for the second fastq file is high, with all averaging in the green (above 28). None of the bases of a mean quality score below 20.

Section 3: Mapping RNA-Seq reads to genome

Q10: Where are most the accepted hits located?

The most accepted hits were located on the gene PSMD3.

Q11: Following Q10, is there any interesting gene around that area?

A gene around the PSMD3 area that I personally find interesting is ORMDL3. This gene also had a significant amount of accepted hits more upstream of PSMD3.

Q12: Cufflinks again produces multiple output files that you can inspect from your right-hand side galaxy history. From the “gene expression” output, what is the FPKM for the ORMDL3 gene? What are the other genes with above zero FPKM values?

The FPKM for ORMDL3 gene is 136853. Other genes with an FPKM above zero include ZPBP2, GSDMA, GSDMB, and PSMD3.